



Violent Extremism and Artificial Intelligence: A Double-Edged Sword in the Context of ASEAN

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Digital integration and the emergence of new technologies such as artificial intelligence (AI) are providing new tools for insurgents to use in spreading their propaganda through violent extremism. The Association of Southeast Asian Nations (ASEAN) has come to represent a conduit for insurgents in planning and carrying out their extreme agendas. This article provides a deeper understanding of the double-edged sword effect of AI in relation to violent extremism in the ASEAN context. It reveals that, even though AI has been very important in countering violent extremism, it has simultaneously facilitated terrorists in spreading their propaganda in more innovative and covert ways. The legal framework governing AI is still in its infancy and challenges such as the double-edged sword effect in the use of the technology require specific guidelines or legislation for use in effective governance.

Biography

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Introduction

Violent extremism and radicalism are not a new phenomenon. They entail diverse beliefs without a specific definition and are not exclusive to any religion, nationality or system of belief (UNDP, 2016). Violent extremism is a broader term than terrorism but encompasses manifestations of terrorism, including ideologically motivated violence (UNODC, 2018). The US defines violent extremism as encouraging, condoning, justifying or supporting the commission of a violent act to achieve political, ideological, religious, social or economic goals (FBI, 2021). The UK Home Office (2023) defines it as vocal and active opposition to fundamental values, which include the rule of law, liberty and mutual respect towards different beliefs and faiths.

Counterterrorism started to evolve with the 9/11 War on Terror, which has had a strong focus on coercive measures, including hard military action, increasing policy powers and expanding intelligence services (CCE, 2023). However, the approach has shifted to become more non-coercive, including the formulation of strategies to prevent individuals from supporting terrorism (ibid.). The United Nations contends that the key elements to countering violent extremism (CVE) involve the use of non-coercive means to drive individuals or groups from using violence and to mitigate recruitment, support, facilitation or engagement (UNODC, n.d.). Neuman (2004) highlighted that CVE strategies involved a non-exhaustive list of activities by governmental and non-governmental entities in the fight to combat radicalisation, such as counter-messaging exercises through social media channels, community engagements, advisory council discussions, capacity-building, women and youth empowerment, and education and training of stakeholders.

In recent years, violent extremism has shifted from an approach of holding specific territories in specific jurisdictions to real-time communications on social media platforms, where those involved seek to spread their ideologies and propaganda.

Comprehensive Analysis of AI-Enhanced Defense Systems in Cyberspace
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 Abstract: The proliferation of cyber threats necessitates innovative defense mechanisms, and this paper presents a comprehensive review and analysis of AI-enabled approaches in cyberspace security. It investigates the integration of artificial intelligence (AI) algorithms, machine learning, and deep learning techniques into cybersecurity frameworks to counteract evolving threats. The study assesses the efficacy of AI in threat detection, anomaly identification, and predictive analysis within diverse cyber environments. Additionally, it scrutinizes the limitations and challenges associated with AI-driven defense strategies, emphasizing the ethical implications, adversarial attacks, and the need for interpretability and transparency in AI models. This research aims to provide a critical evaluation of the current landscape of AI-based defense mechanisms while highlighting their potential in fortifying cyber resilience and addressing emerging cyber threats.
 Key words: threats, defence, AI-driven, cybersecurity
 Introduction: In the contemporary digital landscape, the relentless surge in cyber threats poses an ever-growing challenge to the

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security of information systems and networks. The escalating sophistication and diversity of these threats demand a paradigm shift in defensive strategies. This paper embarks on
nda. The internet has no border control or checks and the lure of anonymity has
changed the characteristics of violent extremism (Jacobsen, 2022; Khodzhanovna,
2023). Terrorist organisations see the internet as a safe way to recruit individuals
and disseminate their ideology, especially through social media platforms such as
Facebook, Twitter, YouTube and Instagram (Broeders et al., 2023).
Meanwhile, emerging technology such as artificial intelligence (AI) has contributed
to the evolution of violent extremism. This article analyses the role of emerging
technologies such as AI in facilitating the commission of acts of terror.

What is violent extremism?

Violent extremism has been a major issue in countries' policies and development
programmes in the past few decades. The term was coined to shift the focus away
from an over-militarised approach after 9/11 and to enable a more moderate
approach in countering and preventing extremism (Saraiva and Erfe, 2023). The
United Nations Plan of Action to Prevent Violent Extremism 2015 aims to develop
resilience in sections of communities that are prone to violent extremism (UNOCT,
2015).

Although the concept is recognised across international communities, a uniform
definition has never been agreed upon. As a result, the concept is easily
manipulated, which poses a critical challenge to authorities and can sometimes also
lead to the over-securitisation of specific sectors to further legitimise the war
against terror (Stephens et al., 2021). The United Nations Special Rapporteur on the
protection of human rights concluded that 'the lack of semantic and conceptual
clarity that surrounds violent extremism remains an obstacle to any in-depth
examination of the impact of strategies and policies to counter violent extremism
on human rights as well as on their effectiveness in reducing the threat of terrorism'
(Emmerson, 2016: para. 55).

Violent extremism includes elements of radicalisation, which is a process of
embracing religious, political and social ideation that causes violent acts between
members or groups (Doosje and van Eerten, 2017; Borum, 2023). Alcalá et al. (2017)
contend that the promotion and adoption of extremist beliefs to advance violence
leads to violent radicalisation, which has a critical effect on religion and society.
Recent research has also highlighted that there are many aspects to extremist



behaviour, emerging from cultural, educational and psychological factors. Interestingly, Stankov et al. (2018) contend that the ideology of extremism and radicalisation depends on mindsets, and extremist behaviours can be found in all humans. This contention supports the concept of extremism immunity, which entails embedding ideas, feelings and behaviours against radicalisation and extremism across all sectors of communities and social categories (ibid.).

Hamin et al. (2021) highlight that violent extremism is multidimensional and very complex, as there is no one agreed definition and different commentators often use the concept interchangeably with terrorism and radicalism. However, despite the lack of a specific meaning, the concept suggests a willingness of individuals or groups to use or support violence.

The proliferation of the internet in the past few decades has also changed how violent extremism is committed. Policy-makers and major stakeholders are strongly aware of the impact of the internet in supporting terrorism and violent extremism. Scrivens et al. (2020) highlight that law enforcement agencies have focused on learning how propaganda and extremist ideas are disseminated and cross into the real world; at the same time, major social media companies are concerned about how their platforms are seen as an important radicalising agent and become the main conduit in promoting real-world violent extremism. Commentators have also described how violent extremists have adopted new digital paradigms in their modus operandi to spread their hateful ideologies and propaganda, recruit new members worldwide and receive funding and tactical support (Salleh et al., 2016; Pressman and Ivan, 2019; Lakomy, 2023). The use of such technologies in violent extremism has been seen in Southeast Asia as well of other parts of the world.

The emergence of artificial intelligence

In the past decade, technology has invaded our everyday lives and dependency has soared. Challenges related to operations and capacity within law enforcement and counterterrorism agencies mean the use of AI has been seen as a holy grail in combatting violent extremism. AI's capacity to process vast amounts of data faster and with greater ease, and to correlate such data and discover patterns and themes, means intelligence agencies see it as an appealing commodity to confront the problem of managing information overload (Bazarkina, 2023). AI can support CVE through automating repetitive tasks, which in turn reduces workloads; predicting



future violent extremist incidence; identifying suspicious transactions to detect terrorism financing; monitoring and moderating content within cyberspace; and other automation of capacities (Gutiérrez-Castillo, 2022). This serves as a game-changer, as all this surpasses human capabilities

Emerging technologies such as AI have been utilised in all sectors, from manufacturing to health to defence. Within the criminal justice context, AI has facilitated investigations through facial recognition; by assisting judges in granting bail and giving out sentences, parole and probation; and in matching DNA to perpetrators (Bazarkina, 2023). Machine learning is used to predict future criminal behaviour and identify patterns and risks of recidivism. AI is also fundamental in the prevention of cybercrime and has proven effective in preventing cyberattacks such as phishing, hacking and terrorism (Garcia, 2019).

Within the context of security, AI has played a crucial part in the fight against terrorism. AI has been used mainly for content moderation since terrorists have taken their operations to the internet (Gunton, 2022). AI and machine learning are believed to have the capacity to reduce terrorist content online and provide a safe place for users to operate within the cyberspace realm (Bamsey and Montasari, 2023). In the Southeast Asian context, such digital transformation requires adaptation, and the increased digitalisation rate exposes the countries to risks, given the established presence of violent extremist groups in the region (Ilyas, 2022).

Generative artificial intelligence

Generative AI is also an issue: these technologies can generate fictional faces and deepfakes. Deepfakes were invented in 2017, as a type of synthetic media that cannot be distinguished from authentic content (Gunton, 2022). Deepfakes are a powerful weapon that violent extremists can utilise, especially in information warfare, when people can no longer rely on what they see and hear online and offline (ibid.). Deepfakes have been used as a tool to commit malicious and criminal activities, usually politically motivated, such as destroying the credibility and reputation of a known individual, harassment, humiliation, extortion and blackmail. This can lead to social unrest and political instability (Bamsey and Montasari, 2023).

Natural language processing

Natural language processing is a deep learning application that analyses a huge amount of natural human language data, reading and defining the meaning in human



languages. The technology involves speech recognition and natural language understanding, generation and translation (Bamsey and Montasari, 2023).

Combatting violent extremism via artificial intelligence

Nations all around the world see AI as a solution to prevent and counter violent extremism. The Council of Europe has adopted the Convention on the Prevention of Terrorism and compiled a Database on Cyberterrorism to mitigate cyberterrorist attacks (Ige et al., 2022). In the UK, steps have been taken to highlight the use of AI in moderating content and providing AI solutions in combatting terrorism online (McKendrick, 2019). Other countries' governments are also confident that the use of AI is the ultimate response to violent extremism.

As well as countries putting in place technology-led solutions to combat extremist content on the internet, major content providers and service providers are creating partnerships to counter violent extremism online (McKendrick, 2019). The Global Internet Forum to Counter Terrorism is an industry-led initiative led by major online platforms such as Microsoft, X (formerly known as Twitter), Facebook and YouTube to combat violent extremism content (Fernandez and Alani, 2021). The aim is to disrupt terrorist activities online and develop tools and capacity to mitigate the impacts of terrorism. AI has also been at the forefront in combatting terrorism: states around the world are using AI in fighting extremist content online (Tech Against Terrorism, 2023). The mechanics of AI, which allow it to analyse and process enormous amounts of data, facilitate law enforcers to track and identify extremist content and activities. Content moderation via AI enables the monitoring and removal of extremist content and the breaking of networks, and the protection of vulnerable individuals who are prone to extremism.

Violent extremism in the Association of Southeast Asian Nations

In the ASEAN context, violent extremism has a long history, dating back to 1948, when the government had to counter new extremist elements during the challenging anti-colonial and post-independence era (El-Muhammady, 2023). The presence of Jemaah Islamiyah, affiliated with Al Qaeda, took root in 1940, and intensified after the 9/11 attacks in New York and Washington, DC. In the 1980s, Indonesia and Malaysia became a platform for violent extremists to participate in multinational jihad against the West. This led Jemaah Islamiyah to set up base in Indonesia, which then became a transnational Southeast Asian terror network committed to a pan-



Southeast Asian Islamic State. Such ideologies spread across the whole region, from southern Thailand across Malaysia into Singapore and Indonesia to the east, and the southern Philippines. This move also led to the Bali bombing of 2002, with more than 200 fatalities. In August 2014, the Malaysian authorities managed to foil planned attacks within the country's capital (Hamzani, 2020).

The Global Terrorism Index, published in 2022, indicates that, of Southeast Asian countries, the Philippines and Myanmar are ranked within the top 20 countries that have been severely impacted by terrorism. Certain online incidents have also led the Indonesian government under the Ministry of Communications and Information Technology to set up a specialised team of moderators to moderate terrorist content in the country (Wilujeng and Risman, 2020). The ASEAN countries (Brunei Darussalam, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, Cambodia and Vietnam) are aware of the growing complexity of violent extremism, especially with the advancement of social media and the internet, resulting in new patterns of radicalisation (ibid.). The major concerns of ASEAN include lone wolf terrorists, regional groups that pledge allegiance to ISIS, Daesh and other terrorist organisations and the return of foreign terrorist fighters to the region (Tay, 2023).

Governing violent extremism in ASEAN

In 2016, the United Nations General Assembly adopted Resolution A/RES/70/291 by consensus to reinforce efforts to fight terrorism and violent extremism. The General Assembly also recommended member states set up regional and national plans of action to be applied within the local context. To align its efforts with those of the rest of the world, ASEAN has put in place and implemented strategies and plans for governance in combatting violent extremism under the ASEAN Comprehensive Plan of Action on Countering Terrorism (Gunaratna, 2018). In 2017, ASEAN adopted the Manila Declaration to Counter the Rise of Radicalisation and Violent Extremism, which includes pledges to implement sustainable and proactive capacity-building, information-sharing between member states, mutual legal assistance on criminal matters and extradition, and the strengthening of mechanisms to address terrorism and violent extremism through collaboration and exchange of experiences with all major stakeholders (Gunaratna, 2018; Habulan et al., 2018).



The ASEAN region has used the soft law approach, given the emphasis on non-interference between the states within the zone (Tan and Nasu, 2016). The ASEAN Convention on Counterterrorism is a framework to govern and co-ordinate member states to adopt a regional treaty on counterterrorism (Shah et al., 2022). Despite the existence of common initiatives throughout the region, Southeast Asian governments have not dealt with terrorism the same way. Indonesia and Singapore adopt a more non-militaristic approach, whereas Malaysia and Thailand rely on more coercive methods.

Malaysia has in place the Prevention of Crime Act 1959, the Prevention of Terrorism Act 2015 and the Special Measures against Terrorism in Foreign Countries Act 2015 to confront the threat of violent extremism by monitoring the activities of foreign terrorist fighters (Hamin et al., 2021). Prior to this, Malaysia had one of the most unpopular pieces of legislation, in the form of the Internal Security Act 1950, which caused citizens to take to the streets to claim that the law violated basic human rights. It was later replaced by the Security Offences (Special Measures) Act (SOSMA) 2012, said to be very similar to the law it replaced (ibid.). SOSMA is a preventive law containing special measures to deal with security-related offences that include terrorism, sabotage and espionage (Dhanapal and Sabaruddin, 2017). The Malaysian Bar Council highlights that the laws in place invoked a low standard of proof and ignored basic safeguards against human rights, leading to numerous civil liberty infringements (ibid.). Despite these controversies, however, the government has set up a Southeast Asian Regional Centre for Counterterrorism, in charge of training, information-sharing and awareness programmes (Hamin et al., 2021).

Similar to the situation in Malaysia, in March 2023 Myanmar published its Anti-Terrorism Bill with the aim of replacing the problematic Prevention of Terrorism Act, which had allegedly led to extensive torture and arbitrary detentions since 1979. The proposed Bill gives the police, the president and the military more broad powers to detain without evidence, prosecute against vaguely defined criminal offences and ban gatherings and organisations. It also does not fulfil the requirements of the United Nations Special Rapporteur, including the need for an appropriate definition of terrorism, the prevention of arbitrary detention, the prohibition of any type of torture, and guaranteed fair trial and due process (Simpson and Farrelly, 2023).



In the Philippines, terrorism is governed under the Human Security Act of 2007. This has a general and broad definition of terrorism that involves elements of fear and panic among the population that coerce the government to give in to unlawful demands (Rasul, 2023). Thailand has its own National Action Plan on terrorism, finalised in 2022, and has criminalised terrorism within its Internal Security Act 2008 (Rasul, 2023). According to Tan and Nasu (2016), Cambodia has a different approach to counterterrorism, given grave concerns about transnational crimes: intel had revealed that Jemaah Islamiyah leaders were freely travelling through the country. This led Cambodia to enact the Law on Counter Terrorism 2007 and the Law on Anti-Money Laundering and Combatting the Financing of Terrorism 2007, to address counterterrorism financing, which was rampant at the time (ibid.). This move started a regional move to address terrorism financing as part of anti-money laundering policies. Malaysia enacted the Anti-Money Laundering, Anti-Terrorism Financing and Proceeds of Unlawful Activities Act 2001 and Myanmar the Control of Money Laundering Law 2002 (Tan and Nasu, 2016; Ramakrishna, 2017); both laws were later amended to include provisions on terrorist financing.

After the Bali bombing, Indonesia enacted the Anti-Terrorism Law 2003, which gives a broad definition of terrorism and allows a suspect to be detained without trial for a period of up to six months (Ramakrishna, 2017). Indonesia, with the help of the USA and Australia, also created Densus 88 in 2003, which serves as a counterterrorism unit to deal with intelligence and operations to dismantle violent extremist networks (Ramakrishna, 2017; Rasul, 2023). Between 2021 and 2023, there were more than 610 people arrested; 42 per cent of them identified as Jemaah Islamiyah members. This shows that the group is still very active in conducting terrorist activities, especially recruitment, fundraising and regeneration, which have now gone online (Subandi et al., 2023).

The double-edged sword of artificial intelligence in violent extremism

Communication interception has been crucial in preventing and countering violent extremism. However, the evolution and increased availability of communication technologies have resulted in these technologies reaching everyone, including sovereign nations, corporations, individuals... and terrorists. It is now easier for terrorists to evade detection, and accessing critical data to predict terrorist attacks has become increasingly challenging for intelligence agencies. Terrorists are also



now implementing more advanced operational security measures in order to evade intelligence collection operations (Bazarkina, 2023). In essence, terrorists have been early adopters of new technologies that have yet to be effectively governed and regulated (Lakomy, 2023). It has been reported that sections of the ISIS terrorist group within ASEAN have also utilised unmanned aerial vehicles or drones to conduct surveillance and reconnaissance (Liang, 2023).

Terrorists have always found ways to adapt and operate in the shadows to ensure non-detection by enforcement officers. The move from real-life terror operations to the online environment is to be expected, given the borderless and anonymous nature of the internet (Brundage et al., 2018). Governments and companies aim to halt the spread of radicalising content by investing in the creation of technologies to counter and identify extremism through AI solutions (ibid.). The main objectives are to understand the phenomena behind online extremism; to detect extreme users and content in cyberspace; and then to predict the spread of extremist ideologies within the online sphere (Bazarkina, 2023).

Amid these herculean efforts to design and develop effective AI solutions to automatically identify and block radical accounts, extremist organisations are also working hard to adapt their behaviour to avoid being detected. Through technological adaption, such organisations can make use of the latest developments in order to increase their reach undetected (Brundage et al., 2018) and can modify the terms and content they post online to avoid being detected by AI technology (UNICRI and UNCCT, 2021). The ASEAN region has always been a destination for terrorist organisations in procuring funding and recruiting new members; Malaysia and Singapore have highlighted that recruitment and funding are being carried out online and that the use of AI makes it challenging to detect such activities (Tay, 2023).

The use of generative AI has also been seen in violent extremists' exploitation of emerging technology. Media spawning involves the use of a single image or video from which generative AI can then generate thousands of manipulated images or videos capable of circumventing automated detection mechanisms utilised by law enforcers. Fully synthetic propaganda generates artificial content, including speeches, images and other propaganda. Personalised propaganda uses tools to customise messaging and media to a targeted audience with specific demographics.



Such technology analyses each demographic and in turn produces personalised propaganda to suit the beliefs and understandings of the audience (Tech Against Terrorism, 2023).

States around the world are using AI to fight extremist content online, and it is seen as a good solution in the fight against online extremism. The mechanics of AI, which allow it to analyse and process enormous amounts of data, facilitate law enforcers to track and identify extremist content and activities. Through content moderation via AI, extremist content is removed and networks are broken (Tech Against Terrorism, 2023). However, any propaganda published is translated into multiple languages via natural language processing software in order to overwhelm moderation. Ultimately, extremist groups have found ways to avoid detection in spreading radicalised content online by subverting moderation, using AI tools to design multiple variants of propaganda specifically engineered to bypass available techniques put in place by law enforcement (Tech Against Terrorism, 2023).

These new emerging technologies have proven that the current model for detection is obsolete, and the use of generative AI will provide opportunities to stay ahead of the threats (Tech Against Terrorism, 2023). Using AI to circumvent safeguards built into the infrastructure amplifies the distribution and dissemination of terror propaganda (Sabbagh, 2023).

The ASEAN response

The current cyber-climate is challenging, with attacks and threats becoming more sophisticated and volatile. The ASEAN regional response to cyberthreats has always been to fortify high levels of co-operation among member states in the form of computer emergency response teams, which focus on capacity-building and information-sharing in cyber-emergencies. However, the region's response to cyberterrorism is fragmented, owing to the lack of a strategic approach towards cybersecurity. In the second quarter of 2023, ASEAN members agreed to develop an ASEAN Guide on AI Governance and Ethics by 2024, following the focus of other nations in AI governance. However, it must be noted that individual ASEAN member states are very slow to progress on having their own AI governance frameworks.

The emergence of generative AI has changed the landscape of cyberterrorism, and ASEAN countries must be prepared. Tay (2023) highlights that ASEAN's current response to cyberterrorism lacks a regional institutional structure, and the cyber-



operation architecture has no clear political authority and is a confusing maze, with various sectoral platforms. The nature of ASEAN itself hinders adequate governance of such crimes; unlike the EU, as a supernational entity, the ASEAN structure is based on intergovernmental organisations and principles of sovereignty, founded in consensus decision-making and non-interference between member states. Apart from this, the uneven development of legal and technological responses to such crimes across ASEAN is seen as a limitation.

Tay (2023) has also contended that the deficiency of the common cyber-lexicon is also a challenge: different states have different measures in defining the impact of a cyber-emergency or an attack on critical national infrastructure. Malaysia, for example defines a Level 5 crisis as having a critical impact on critical infrastructure organisations; however, other countries, such as Cambodia and Indonesia, do not have similar responses to Malaysia in defining a crisis.

Conclusion

There has been unprecedented progress in the use of AI in countering cyberterrorism to predict and detect terrorist attacks. However, significant challenges also arise with the deployment of such technologies. ASEAN should put in place a standard and agreed terminology to ensure effective communication and information-sharing during a cyber-emergency such as a cyberterrorism attack. The absence of such an agreement will have impacts on the counter-effort to eliminate terrorist threats within the region. Member states within ASEAN should also ensure that the datasets used in countering terrorist attacks are verified and free from the risks of hallucinations and data poisoning. Such risks are aggravated given that the majority of states within ASEAN do not yet have a framework in place to govern AI and are still in the planning stages on addressing the issue. Meanwhile, despite ASEAN's ongoing participation in the effort to ensure global governance in AI, national regulation and regional co-operation are still lacking.

Technology will always evolve and the double-edged impact of AI in cyberterrorism means states must be prepared to face the unpredictable risks associated with it. The constant evolution of extremist behaviours and the numerous ways of avoiding detection in ever-changing narratives call for a serious response within the context of ASEAN. The fight to diffuse the impacts of negative uses of AI technology will continue. Enforcement is the key. Having a clear legal framework to combat



terrorism and AI governance is of the utmost importance. Apart from this, digital citizens must be made aware of the subversive nature of the internet and social media, and national action plans and strategies must be put in place to prevent violent extremism.

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Edexcel Single Chemistry (1C10) from 2016 Topic C1a&b				
Topic	Student Checklist	R	A	G
Topic 1a – Key concepts in chemistry Edexcel Single Chemistry (1C10) from 2016 Topic C1a&b	Describe how the Dalton model of an atom has changed over time because of the discovery of subatomic particles	-	-	-
	Describe the structure of an atom as a nucleus containing protons and neutrons, surrounded by electrons in shells	-	-	-
	Recall the relative charge and relative mass of: a proton, a neutron and an electron	-	-	-
	Explain why atoms contain equal numbers of protons and electrons	-	-	-
	Describe the nucleus of an atom as very small compared to the overall size of the atom	-	-	-
	Recall that most of the mass of an atom is concentrated in the nucleus	-	-	-
	Recall the meaning of the term mass number of an atom	-	-	-
	Describe atoms of a given element as having the same number of protons in the nucleus and that this number is unique	-	-	-
	Describe what isotopes are	-	-	-
	Calculate the numbers of protons, neutrons and electrons in atoms given the atomic number and mass number	-	-	-
	Explain how the existence of isotopes results in relative atomic masses of some elements not being whole numbers	-	-	-
	HT-ONLY: Calculate the relative atomic mass of an element from the relative masses and abundances of its isotopes	-	-	-
	Describe how Mendeleev arranged the elements known at that time, in a periodic table by using properties of these elements and their compounds	-	-	-
	Describe how Mendeleev used his table to predict the existence and properties of some elements not discovered by then	-	-	-
	Explain that Mendeleev thought he had arranged elements in order of increasing relative atomic mass but this was not always true	-	-	-
	Explain the meaning of atomic number of an element in terms of position in the periodic table and number of protons in the nucleus	-	-	-
	Describe how elements are arranged in the groups and periods of the periodic table	-	-	-
	Identify elements as metals or non-metals according to their position in the periodic table, explaining this division in terms of atomic structure	-	-	-
	Predict the electronic configurations of the first 20 elements in the periodic table as diagrams and in the form 2.8.1 etc	-	-	-
	Explain how the electronic configuration of an element is related to its position in the periodic table	-	-	-
	Explain how ionic bonds are formed to produce cations and anions, including the use of dot and cross diagrams	-	-	-
	Recall that an ion is an atom or group of atoms with a positive or negative charge	-	-	-
	Calculate the numbers of protons, neutrons and electrons in simple ions given the atomic number and mass number	-	-	-
	Explain the formation of ions in ionic compounds from their atoms, limited to compounds of elements in groups 1, 2, 6 and 7	-	-	-
	Explain the use of the endings -ide and -ate in the names of compounds	-	-	-
	Deduce the formulae of ionic compounds given the formulae of the constituent ions	-	-	-
	Explain the structure of an ionic compound including a description of the lattice and electrostatic forces	-	-	-



Topic 1b – Key concepts in chemistry	Explain how a covalent bond is formed when a pair of electrons is shared between two atoms	-	-	-
	Recall that covalent bonding results in the formation of molecules	-	-	-
	Recall the typical size (order of magnitude) of atoms and small molecules	-	-	-
	Explain the formation of simple molecular, covalent substances, using dot and cross diagrams, including: H , HCl , H_2O , CH_4 , O_2 , CO_2	-	-	-
	Explain why elements and compounds can be classified as: ionic, simple molecular (covalent), giant covalent and metallic	-	-	-
	Explain how the structure and bonding of substances results in different physical properties	-	-	-
	Explain the properties of ionic compounds limited to: melting/boiling points, forces between ions and conductivity	-	-	-
	Explain the properties of typical covalent, simple molecular compounds limited to: melting/boiling points, forces between ions and conductivity	-	-	-
	Recall that graphite and diamond are different forms of carbon and that they are examples of giant covalent substances	-	-	-
	Describe the structures of graphite and diamond	-	-	-
	Explain, in terms of structure and bonding, why graphite and diamond have different uses	-	-	-
	Explain the properties of fullerenes including C_{60} and graphene in terms of their structures and bonding	-	-	-
	Describe, using poly(ethene) as the example, that simple polymers consist of large molecules containing chains of carbon atoms	-	-	-
	Explain the properties of metals, including malleability and the ability to conduct electricity	-	-	-
	Describe the limitations of particular representations and models, to include dot & cross, ball & stick models & 2/3D	-	-	-
	Describe the properties of most metals	-	-	-
	Calculate relative formula mass given relative atomic masses	-	-	-
	Calculate the formulae of simple compounds from reacting masses and understand that these are empirical formulae	-	-	-
	Deduce: empirical formula of a compound from the formula of its molecule	-	-	-
	Deduce: molecular formula of a compound from its empirical formula and its relative molecular mass	-	-	-
	Describe an experiment to determine the empirical formula of a simple compound such as magnesium oxide	-	-	-
	Explain the law of conservation of mass applied to: a closed system and a non-enclosed system	-	-	-
	Calculate masses of reactants and products from balanced equations, given the mass of one substance	-	-	-
	Calculate the concentration of solutions in $g\ dm^{-3}$	-	-	-
	HT ONLY: Recall what one mole of particles of a substance is defined as	-	-	-
	HT ONLY: Calculate the number of: moles of particles of a substance in a given mass of that substance and vice versa	-	-	-
	HT ONLY: Calculate the number of: particles of a substance in a given number of moles of that substance and vice versa	-	-	-
	HT ONLY: Calculate the number of: particles of a substance in a given mass of that substance and vice versa	-	-	-
	HT ONLY: Explain why, in a reaction, the mass of product formed is controlled by the mass of the reactant which is not in excess	-	-	-
	HT ONLY: Deduce the stoichiometry of a reaction from the masses of the reactants and products	-	-	-



Edexcel Single Chemistry (1CI0) from 2016 Topics C6&7				
Topic	Student Checklist	R	A	G
Topic 6 – Groups in the periodic table	Explain why some elements can be classified as alkali metals, halogens or noble gases, based on their position in the periodic table	-	-	-
	Recall the physical properties of alkali metals	-	-	-
	Describe the reactions of lithium, sodium and potassium with water	-	-	-
	Describe the pattern in reactivity of the alkali metals, lithium, sodium and potassium, with water, and use this pattern to predict the reactivity of other alkali metals	-	-	-
	Explain this pattern in reactivity in terms of electronic configurations	-	-	-
	Recall the colours and physical states of chlorine, bromine and iodine at room temperature	-	-	-
	Describe the pattern in the physical properties of the halogens, chlorine, bromine and iodine, and use this pattern to predict the physical properties of other halogens	-	-	-
	Describe the chemical test for chlorine	-	-	-
	Describe the reactions of the halogens, chlorine, bromine and iodine, with metals to form metal halides, and use this pattern to predict the reactions of other halogens	-	-	-
	Recall that the halogens, chlorine, bromine and iodine, form hydrogen halides which dissolve in water to form acidic solutions, and use this pattern to predict the reactions of other halogens	-	-	-
	Describe the relative reactivity of the halogens chlorine, bromine and iodine, as shown by their displacement reactions with halide ions and use this to predict the reactions of astatine	-	-	-
	HT ONLY: Explain why these displacement reactions are redox reactions in terms of gain and loss of electrons, identifying which of these are oxidised and which are reduced	-	-	-
	Explain the relative reactivity of the halogens in terms of electronic configurations	-	-	-
	Explain why the noble gases are chemically inert, compared with the other elements, in terms of their electronic configurations	-	-	-
	Explain how the uses of noble gases depend on their inertness, low density and/or non-flammability	-	-	-
	Describe the pattern in the physical properties of some noble gases and use this pattern to predict the physical properties of other noble gases	-	-	-
Topic 7 – Rates of reaction and energy changes	<i>Core Practical: Investigate the effects of changing the conditions of a reaction on the rates of chemical reactions by: measuring the production of a gas/observing a colour change</i>	-	-	-
	Suggest practical methods for determining the rate of a given reaction	-	-	-
	Explain how reactions occur by discussing the collision theory	-	-	-
	Explain the effects on rates of reaction of changes in temperature, concentration, surface area to volume ratio and pressure in terms of frequency and energy of collisions	-	-	-
	Interpret graphs of mass, volume or concentration of reactant or product against time	-	-	-
	Describe what a catalyst is	-	-	-
	Explain how the addition of a catalyst increases the rate of a reaction in terms of activation energy	-	-	-
	Recall that enzymes are biological catalysts and that enzymes are used in the production of alcoholic drinks	-	-	-
	Recall when chemical changes occur that they cause changes in heat energy	-	-	-
	Describe the differences between endothermic and exothermic in terms of energy taken in or given out	-	-	-
	Recall if bonds are broken or made for each of the following reactions: endothermic and exothermic	-	-	-
	Describe why the overall heat energy change for a reaction is exothermic or endothermic in terms of bonds being made or broken	-	-	-
	HT ONLY: Calculate the energy change in a reaction given the energies of bonds (in kJ mol⁻¹)	-	-	-



	Explain the term activation energy	-	-	-
	Draw and label reaction profiles for endothermic and exothermic reactions, identifying activation energy	-	-	-



Edexcel Single Chemistry (1C10) from 2016 Topic C8				
Topic	Student Checklist	R	A	G
Topic 8 – Fuels and Earth science	Recall what a hydrocarbon is	-	-	-
	Describe and explain what crude oil is and why it is important	-	-	-
	Describe and explain the separation of crude oil into simpler, more useful mixtures by the process of fractional distillation	-	-	-
	Recall the names and uses of the following fractions: gases, petrol, kerosene, diesel oil, fuel oil and bitumen	-	-	-
	Explain how hydrocarbons in different fractions differ from each other in terms of boiling point, number of C & H's, flammability and viscosity	-	-	-
	Explain what a homologous series of hydrocarbon compounds is	-	-	-
	Describe the complete combustion of hydrocarbon fuels including energy changes and products	-	-	-
	Explain why the incomplete combustion of hydrocarbons can produce carbon and carbon monoxide	-	-	-
	Explain how carbon monoxide behaves as a toxic gas	-	-	-
	Describe the problems caused by incomplete combustion in appliances that use carbon compounds as fuels	-	-	-
	Explain how impurities in some hydrocarbon fuels result in the production of sulfur dioxide	-	-	-
	Explain some problems associated with acid rain	-	-	-
	Explain why, when fuels are burned in engines, oxides of nitrogen are formed and that they are pollutants	-	-	-
	Evaluate the advantages and disadvantages of using hydrogen, rather than petrol, as a fuel in cars	-	-	-
	Recall the names and sources of some renewable fossil fuels	-	-	-
	Explain what cracking is and why it is necessary	-	-	-
	Recall that the gases produced by volcanic activity formed the Earth's early atmosphere	-	-	-
	Describe what the Earth's early atmosphere was thought to contain	-	-	-
	Explain what the oceans were formed from	-	-	-
	Explain why the amount of carbon dioxide in the atmosphere decreases when the oceans were formed	-	-	-
	Explain how the growth of primitive plants changes the composition of gases in the atmosphere	-	-	-
	Describe the chemical test for oxygen	-	-	-
	Describe and explain the greenhouse effect and name the gases that contribute to it	-	-	-
	Evaluate the evidence for human activity causing climate change	-	-	-
	Describe the potential effects on the climate of increased levels of carbon dioxide and methane generated by human activity	-	-	-
	Describe how effects on the climate may be mitigated: consider scale, risk and environmental implications	-	-	-



Edexcel Single Chemistry (1C10) from 2016 Topic C9a&b (Chem only)				
Topic	Student Checklist	R	A	G
Topic 9a – Separate chemistry 2 parts a	Chem ONLY: Explain why the test for any ion must be unique	-	-	-
	Chem ONLY: Describe flame tests to identify the following ions in solids: Li^+ , Na^+ , K^+ , Ca^{2+} , Cu^{2+} including the colours of the flames	-	-	-
	Chem ONLY: Describe tests to identify the following ions: Al^{3+} , Ca^{2+} , Cu^{2+} , Fe^{2+} , Fe^{3+} , NH_4^+ using NaOH solution	-	-	-
	Chem ONLY: Describe the chemical test for ammonia	-	-	-
	Chem ONLY: Describe tests to identify the following ions: CO_3^{2-} , SO_4^{2-} , Cl^- , Br^- , I^-	-	-	-
	Chem ONLY: Core Practical: Identify the ions in unknown salts, using the tests for the specified cations and anions in the specification	-	-	-
	Chem ONLY: Identify the ions in unknown salts, using results of the tests stated	-	-	-
	Chem ONLY: Describe that instrumental methods of analysis are available and that these may improve sensitivity, accuracy and speed of tests	-	-	-
	Chem ONLY: Evaluate data from a flame photometer to determine the concentration of ions in dilute solution using a calibration curve	-	-	-
	Chem ONLY: Evaluate data from a flame photometer to identify metal ions by comparing the data with reference data	-	-	-
	Chem ONLY: To identify metal ions by comparing the data with reference data	-	-	-
	Chem ONLY: Recall the formulae of molecules of the alkanes, methane, ethane, propane and butane, and draw the structure of these	-	-	-
	Chem ONLY: Explain why the alkanes are saturated hydrocarbons	-	-	-
	Chem ONLY: Recall the formulae of molecules of the alkenes, ethene, propene, butene, and draw the structures (but 1-ene and but 2-ene only)	-	-	-
	Chem ONLY: Explain why the alkenes are unsaturated hydrocarbons	-	-	-
	Chem ONLY: Recall the addition reaction of ethene with bromine, showing the structures of reactants and products, and extend this to other alkenes	-	-	-
	Chem ONLY: Explain how bromine water is used to distinguish between alkanes and alkenes	-	-	-
	Chem ONLY: Describe how the complete combustion of alkanes and alkenes involves the oxidation of the hydrocarbons, name the products	-	-	-



Topic 9b – Separate chemistry 2 part b	Chem ONLY: Recall that a polymer is a substance of high average relative molecular mass made up of small repeating units	-	-	-
	Chem ONLY: Describe how ethene molecules can combine together in a polymerisation reaction	-	-	-
	Chem ONLY: Describe that the addition polymer formed is called polyethene	-	-	-
	Chem ONLY: Describe how other addition polymers can be made by combining together other monomer molecules containing C=C	-	-	-
	Chem ONLY: Describe how to deduce the structure of a monomer from the structure of an addition polymer and vice versa	-	-	-
	Chem ONLY: Explain how the uses of polymers are related to their properties and vice versa	-	-	-
	HT & Chem ONLY: Explain why polyesters are condensation polymers	-	-	-
	HT & Chem ONLY: Explain how a polyester is formed when a monomer molecule containing two carboxylic acid groups is reacted with a monomer molecule containing two alcohol groups	-	-	-
	HT & Chem ONLY: Explain how a molecule of water is formed each time an ester link is formed	-	-	-
	Chem ONLY: Describe some problems associated with polymers including the availability of starting materials	-	-	-
	Chem ONLY: Describe the advantages and disadvantages of recycling polymers, including economic implications, availability of starting materials and environmental impact	-	-	-
	Chem ONLY: Recall that DNA is a polymer made from four different monomers called nucleotides	-	-	-
	Chem ONLY: Recall that starch is a polymer based on sugars	-	-	-
	Chem ONLY: Recall that proteins are polymers based on amino acids	-	-	-
	Chem ONLY: Recall the formulae of molecules of the alcohols, methanol, ethanol, propanol and butanol, and draw the structures of these molecules, showing all covalent bonds	-	-	-
	Chem ONLY: Recall that the functional group in alcohols is –OH	-	-	-
	Chem ONLY: Core Practical: Investigate the temperature rise produced in a known mass of water by the combustion of the alcohols ethanol, propanol, butanol and pentanol	-	-	-
	Chem ONLY: Recall the formulae of molecules of the carboxylic acids, methanoic, ethanoic, propanoic and butanoic acids, and draw the structures of these molecules, showing all covalent bonds	-	-	-
	Chem ONLY: Recall that the functional group in carboxylic acids is –COOH	-	-	-
	Chem ONLY: Recall that ethanol can be oxidised to produce ethanoic acid and extend this to other alcohols	-	-	-
	Chem ONLY: Recall members of a given homologous series have similar reactions because their molecules contain the same functional group and use this to predict the products of other in series	-	-	-
	Chem ONLY: Describe the production of ethanol by fermentation of carbohydrates in aqueous solution, using yeast to provide enzymes	-	-	-
	Chem ONLY: Explain how to obtain a concentrated solution of ethanol by fractional distillation of the fermentation mixture	-	-	-
	Chem ONLY: Compare the size of nanoparticles with the sizes of atoms and molecules	-	-	-
	Chem ONLY: Describe how the properties of nanoparticulate materials are related to their uses including surface area to volume ratio of the particles they contain, including sunscreens	-	-	-
	Chem ONLY: Explain the possible risks associated with some nanoparticulate materials	-	-	-
	Chem ONLY: Compare, using data, the physical properties of glass and clay ceramics, polymers, composites and metals	-	-	-
	Chem ONLY: Describe how the properties of nanoparticulate materials are related to their uses including surface area to volume ratio of the particles they contain, including sunscreens	-	-	-

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Chem ONLY: Explain the possible risks associated with some nanoparticulate materials	-	-	-
Chem ONLY: Compare, using data, the physical properties of glass and clay ceramics, polymers, composites and metals	-	-	-
Chem ONLY: Explain why the properties of a material make it suitable for a given use and use data to select materials appropriate for specific uses	-	-	-